

Learning Intention: I can collect data to answer a question and record it using tallies and a frequency table.

Success Criteria — I can:

- Ask a clear question that can be answered by collecting data.
- Use tally marks to record data as I collect it.
- Organise tally data into a frequency table.

Strategy Box — Four Steps to Collecting Data

1) Ask a clear question (e.g. "What is your favourite season?"). **2)** Decide the categories (e.g. Summer, Autumn, Winter, Spring). **3)** Collect responses, marking a tally each time. **4)** Count the tallies into a frequency table.

Worked Example — Favourite Season Survey

Season	Tally	Frequency
Summer		11
Autumn		4
Winter		6
Spring		9

Each single line is one response. A group of 5 is shown as 4 upright lines with one line crossing through — this makes counting large totals quick and accurate.

Review: Reading Tallies

Count the crossed groups of 5 first, then add any leftover single lines. E.g. 2 groups of 5 plus 3 leftover = 13.

Try It

How many responses does this tally show? 

Answer: _____

Parent Tip

Run a mini survey at home — ask family members a question like "What's your favourite fruit?" and record answers together using tally marks. Count up the totals when you're done.

Key Vocabulary

Word	Meaning
Tally	A quick mark used to record each response as data is collected
Frequency	The total number of times something occurs
Category	A group that responses can be sorted into

Build Skills — Record and organise data.**Q1 — Read the Tally Chart: Favourite Pet**

Pet	Tally	Frequency
Dog		
Cat		
Fish		
Bird		

a) Fill in the Frequency column. b) Which pet got the most votes? _____ c) What is the total number of votes? _____

Q2 — Record the Data

Ten students were asked their favourite school subject. Here are the responses in order: Maths, Art, Maths, Sport, Art, Maths, Sport, Art, Maths, Sport. Record each response as a tally, then write the frequency.

Subject	Tally	Frequency
Maths		
Art		
Sport		

Q3 — Design Your Own Survey Question

Write a survey question you could ask your classmates, and list 3–4 categories for the answers.

Question: _____

Category	Category	Category	Category

Watch Out!

Always mark one tally per response, in the order the data comes in — don't skip or double-count. When you reach the 5th tally in a group, cross through the previous 4 to make counting easier later.

Q4 — Frequency Table to Tally

A frequency table shows Red = 8, Blue = 3, Green = 6. Draw the matching tally marks for each colour.

Colour	Frequency	Draw the tally
Red	8	
Blue	3	
Green	6	

Apply & Reasoning — Use what you know to explain your thinking.

- 1 Reasoning:** Why is it better to use tally marks while collecting data, rather than trying to remember the count in your head?

- 2 Reasoning:** A tally shows 3 full groups of 5 and 2 leftover lines. What is the frequency? Explain how you worked it out.

- 3 Open-ended:** Plan a data collection: write a question of interest, list your categories, and describe exactly how you would collect the data from your class (who you'd ask, how you'd record it).

Quick Check — Exit Ticket

Draw a tally for 8. How many groups of 5 did you need?

Answer & Teacher Support

Answer Key — Pages 1 & 2

Try It (p1): The sample tally shown = 7.

Q1 (p2): Dog=9, Cat=7, Fish=3, Bird=5 (matching the tallies drawn). b) Dog got the most votes. c) Total = $9+7+3+5 = 24$.

Q2 (p2): Maths: 4 tallies, frequency 4. Art: 3 tallies, frequency 3. Sport: 3 tallies, frequency 3. (Total = 10, matching the 10 responses given.)

Q3 (p2): Accept any clear survey question with 3–4 sensible, non-overlapping categories.

Q4 (p2): Red: 1 group of 5 (crossed) + 3 = 8. Blue: 3 single lines. Green: 1 group of 5 (crossed) + 1 = 6.

Answer Key — Page 3

Q1: Tallies are quick to mark in the moment and far less likely to lose count than trying to remember a running total mentally, especially with many responses.

Q2: $3 \times 5 = 15$, plus 2 leftover = 17.

Q3 (acceptable-answer guide): Accept any complete plan with a clear question, sensible categories, and a realistic, specific method for collecting responses (e.g. asking each classmate individually and marking a tally sheet).

Exit ticket: A tally for 8 needs 1 full group of 5 (crossed) plus 3 single lines.

Teacher Checklist

- ☐ Reads tally marks accurately, including groups of 5.
- ☐ Records raw data as tally marks correctly.
- ☐ Converts tallies into a frequency table.
- ☐ Writes a clear, answerable survey question with sensible categories.
- ☐ Draws tally marks correctly from a given frequency.
- ☐ Describes a realistic data collection method.

Teacher Tip

Watch for students who lose track of the "gate" pattern (4 lines + 1 diagonal) and instead draw uneven groups — model this explicitly before independent practice. This worksheet pairs well with the "Reading & Creating Graphs" worksheet (AC9M3ST02) as a natural next step.